AI 321: Machine Learning for Sequential Data

Introduction to z-Transforms

Rhandley D. Cajote, PhD

Electrical and Electronics Engineering Institute (EEEI)
University of the Philippines Diliman

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Overview

1. What is a z-transform?

2. Why is it important?

Definition: z-Transform

- Direct z-transform of a DT signal $x(n)$ is defined as a power series:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \tag{1}$$

  where z is a complex variable

- Notation:

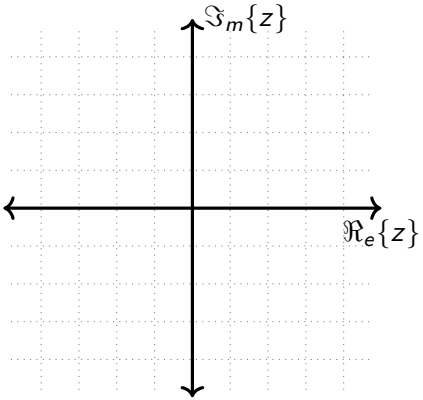
$$X(z) = Z\{x(n)\}, \qquad x(n) \overset{z}{\Longleftrightarrow} X(z)$$

- Transforms the discrete-time signal $x(n)$ into its complex plane representation, $X(z)$
- Enables analysis of discrete-time signals and LTI systems (similar to Laplace for CT)

Z-plane

We can visualize $X(z)$ using the z-plane (complex).

$\Re_e\{z\} \rightarrow$ x-axis, $\Im_m\{z\} \rightarrow$ y-axis



Region of Convergence (ROC) of $X(z)$

Region of Convergence (ROC)

The **Region of Convergence** (ROC) is the set of all value of z for which $X(z)$ attains a finite value

The ROC should always be be indicated whenever the z-transform $X(z)$ is determined.

Region of Convergence of $X(z)$: Examples

1. If $x(n) = \{\underset{\uparrow}{1}, 2, 5, 7, 0, 1\}$, what is $X(z)$ and ROC?

   From the definition (1),

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} \\ &= x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} \\ &\quad + x(4)z^{-4} + x(5)z^{-5} \\ &= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5} \end{aligned}$$

   ROC: entire z-plane, except at $z = 0$

Region of Convergence of $X(z)$: Examples

2. $x(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$X(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$$

ROC: entire z-plane except $z = 0$ and $z = \infty$.

3. $x(n) = \delta(n) = \{1\}$

$$X(z) = 1, \quad \text{ROC: entire } z\text{-plane}$$

Region of Convergence of $X(z)$: Examples

4. $x(n) = (\frac{1}{2})^n u(n), \quad x(n) = \{\underset{\uparrow}{1}, \frac{1}{2}, (\frac{1}{2})^2, \ldots, (\frac{1}{2})^k, \ldots\}$

   Using the definition, (1)

$$X(z) = \sum_{n=0}^{\infty} (\tfrac{1}{2})^n z^{-n}$$
$$= \sum_{n=0}^{\infty} \left(\tfrac{1}{2} z^{-1}\right)^n = 1 + \tfrac{1}{2} z^{-1} + (\tfrac{1}{2})^2 z^{-2} + (\tfrac{1}{2})^3 z^{-3} + \ldots$$

   This is a geometric series: $1 + A + A^2 + A^3 + \ldots = \frac{1}{1-A}$, if $A < 1$

   Hence,

$$X(z) = \frac{1}{1 - \frac{1}{2} z^{-1}} \qquad \text{ROC} : |z| > \frac{1}{2}$$

Impulse response and System function

- If the system is described by the impulse response, $h(n)$, then we have:

$$x(n) \longrightarrow \boxed{h(n)} \longrightarrow y(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} h(n-k)x(k)$$

- The z-transform of the system is described by, $H(z)$, system function.

$$X(z) \longrightarrow \boxed{H(z)} \longrightarrow Y(z)$$

$$Y(z) = H(z)X(z), \qquad H(z) = \frac{Y(z)}{X(z)}$$

Insights about $H(z)$

$H(z)$	Insight
$H(z) = a0 + a_1 z^{-1} + a_2 z^{-2}$	MA(2) process
$H(z) = 1/(b0 + b_1 z^{-1} + b_2 z^{-2})$	AR(2) process
$H(z) = \frac{a0 + a_1 z^{-1} + a_2 z^{-2}}{b0 + b_1 z^{-1} + b_2 z^{-2}}$	ARMA(2,2) process
$H(z) = 1/(1 - \frac{3}{2} z^{-1})$	Unstable system (BIBO)
$H(z) = 1 - \frac{3}{2} z^{-1}$	Stable system (BIBO)
$H(z) = 1/(1 - \frac{2}{3} z^{-1})$	simple low-pass filter (stable)
$H(z) = 1/(1 + \frac{2}{3} z^{-1})$	simple high-pass filter (stable)

more insights about $H(z)$

$H(z)$	Insight
$H(z) = 1 - z^{-2}$	simple bandpass filter with zero DC offset
$H(z) = 1/(1 + \frac{81}{100}z^{-2})$	bandpass filter (high-Q)
$H(z) = 1 + \frac{81}{100}z^{-2}$	bandstop filter
$H(z) = 1/(1 + z^{-2})$	oscillator (bi-stable)

more insights

- $H(z) = (1 - z^{-4})$, linear phase comb filter

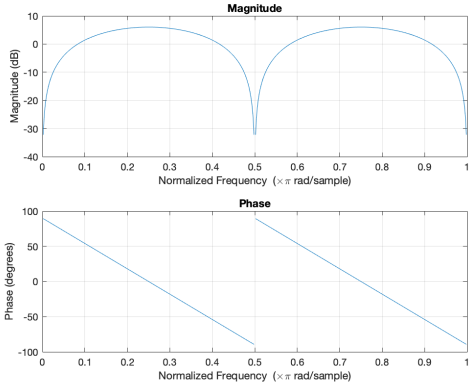


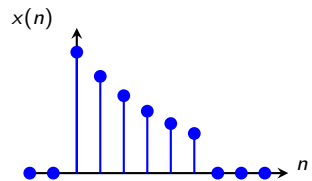
Figure: Frequency response of the comb filter

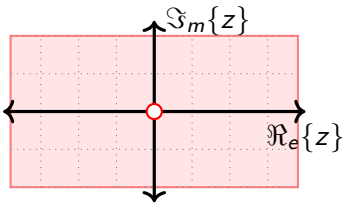
In Matlab, type: figure; freqz([1 0 0 0 -1],1);

Finite Duration Signals ROC

Causal Signals

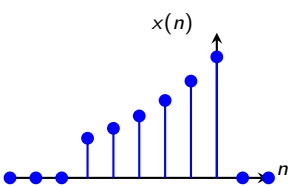
ROC: entire z-plane except $z = 0$

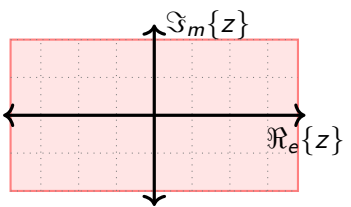




Anti-Causal Signals

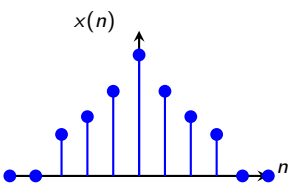
ROC: entire z-plane except $z = \infty$



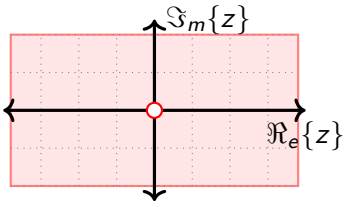


Finite Duration Signals ROC

Two-Sided Signals

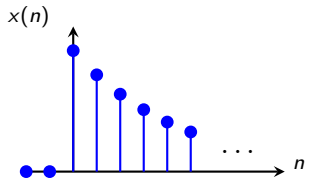


ROC: entire z-plane
except $z = 0$ and $z = \infty$

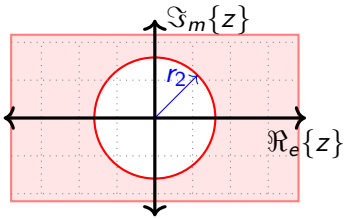


Infinite Duration Signals ROC

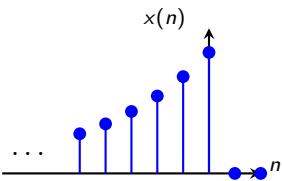
Causal Signals



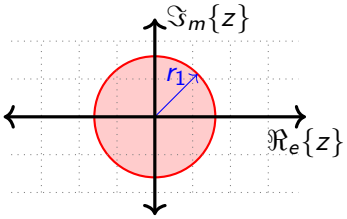
ROC: $|z| > r_2$



Anti-Causal Signals

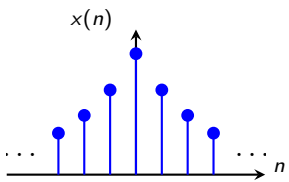


ROC: $|z| < r_1$

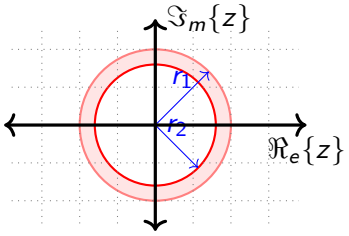


Infinite Duration Signals ROC

Two-Sided Signals



ROC: $r_2 < |z| < r_1$



Causality and the z-Transform

- Causal systems have ROCs that are outside a circle
- Anticausal systems have ROCs that are inside of a circle
- Non-causal systems have ROCs that form annular regions.

Properties of the z-Transform

1. **Linearity**.
   If
   $$x_1(n) \overset{z}{\Longleftrightarrow} X_1(z) \quad \text{and} \quad x_2(n) \overset{z}{\Longleftrightarrow} X_2(z)$$
   then
   $$x(n) = a_1 x_1(n) + a_2 x_2(n) \overset{z}{\Longleftrightarrow} a_1 X_1(z) + a_2 X_2(z)$$
   for any constant a_1 and a_2.

Example 1.A: Linearity

Determine the z-transform and ROC of the signal:

$$x(n) = [3(2^n) - 4(3^n)]u(n)$$

Solution: Define $x_1(n) = 2^n u(n)$ and $x_2(n) = 3^n u(n)$ then,

$$x(n) = 3x_1(n) - 4x_2(n)$$

by linearity the z-transform is

$$X(z) = 3X_1(z) - 4X_2(z)$$

We know that

$$\alpha^n u(n) = \frac{1}{1 - \alpha^n z^{-1}}, \quad \text{ROC: } |z| > |\alpha|$$

Example 1.A: Linearity (continued)

By setting $\alpha = 2$ and $\alpha = 3$, we obtain

$$x_1(n) = 2^n u(n) \overset{z}{\Longleftrightarrow} X_1(z) = \frac{1}{1 - 2z^{-1}} \quad \text{ROC: } |z| > 2$$

$$x_2(n) = 3^n u(n) \overset{z}{\Longleftrightarrow} X_2(z) = \frac{1}{1 - 3z^{-1}} \quad \text{ROC: } |z| > 3$$

The intersection of the ROC of $X_1(z)$ and $X_2(z)$ is $|z| > 3$, thus the overall z-transform

$$X(z) = \frac{3}{1 - 2z^{-1}} - \frac{4}{1 - 3z^{-1}} \quad \text{ROC: } |z| > 3$$

Note: When combining several z-transforms, the ROC of the overall transform is at least the intersection of the ROCs of the individual transforms. Except when the signal combination results in a finite-signal.

Example 1.B: Linearity

Determine the z-transform of the signals:

a. $x(n) = (\cos \omega_0 n)u(n)$
b. $x(n) = (\sin \omega_0 n)u(n)$

a. Solution: Using Euler's identity

$$x(n) = (\cos \omega_0 n)u(n) = \frac{1}{2}e^{j\omega_0 n}u(n) + \frac{1}{2}e^{-j\omega_0 n}u(n)$$

Thus, by linearity

$$X(z) = \frac{1}{2}Z\{e^{j\omega_0 n}u(n)\} + \frac{1}{2}Z\{e^{-j\omega_0 n}u(n)\}$$

If we set $\alpha = e^{\pm j\omega_0}$ ($|\alpha| = |e^{\pm j\omega_0}| = 1$), we have

$$e^{j\omega_0 n}u(n) \overset{z}{\Longleftrightarrow} \frac{1}{1 - e^{j\omega_0}z^{-1}} \quad \text{ROC: } |z| > 1$$

Example 1.B: Linearity (continued)

a. Solution: (continued)

If we set $\alpha = e^{\pm j\omega_0}$ ($|\alpha| = |e^{\pm j\omega_0}| = 1$), we also have

$$e^{-j\omega_0 n}u(n) \overset{z}{\Longleftrightarrow} \frac{1}{1 - e^{-j\omega_0}z^{-1}} \quad \text{ROC: } |z| > 1$$

Thus,

$$X(z) = \frac{1}{2}\frac{1}{(1 - e^{j\omega_0}z^{-1})} + \frac{1}{2}\frac{1}{(1 - e^{-j\omega_0}z^{-1})} \quad \text{ROC: } |z| > 1$$

Combining terms, we have

$$\cos(\omega_0 n)u(n) \overset{z}{\Longleftrightarrow} \frac{1 - z^{-1}\cos(\omega_0)}{1 - 2z^{-1}\cos(\omega_0) + z^{-2}} \quad \text{ROC: } |z| > 1$$

Example 1.B: Linearity (continued)

b. Solution: Using Euler's identity

$$x(n) = (\sin \omega_0 n)u(n) = \frac{1}{2j}\left(e^{j\omega_0 n}u(n) - e^{-j\omega_0 n}u(n)\right)$$

Thus,

$$X(z) = \frac{1}{2j}\left(\frac{1}{(1 - e^{j\omega_0}z^{-1})} - \frac{1}{(1 - e^{-j\omega_0}z^{-1})}\right) \quad \text{ROC: } |z| > 1$$

and finally

$$\sin(\omega_0 n)u(n) \overset{z}{\Longleftrightarrow} \frac{z^{-1}\sin(\omega_0)}{1 - 2z^{-1}\cos(\omega_0) + z^{-2}} \quad \text{ROC: } |z| > 1$$

Properties of the z-Transform

2. **Time-shifting**.
   If
   $$x_1(n) \overset{z}{\Longleftrightarrow} X_1(z)$$
   then
   $$x(n-k) \overset{z}{\Longleftrightarrow} z^{-k}X(z)$$
   ROC of $z^{-k}X(z)$ is the same as $X(z)$ except for $z = 0$ if $k > 0$ and for $z = \infty$ if $k < 0$.

   The proof of the property follows directly from the definition.

Example 2: Time-shifting

Given the signals:

$$x_1(n) = [\underset{\uparrow}{1}, 2, 5, 7, 0, 1]$$
$$x_2(n) = [1, 2, \underset{\uparrow}{5}, 7, 0, 1]$$
$$x_3(n) = [\underset{\uparrow}{0}, 0, 1, 2, 5, 7, 0, 1]$$

Determine the z-transform of $x_2(n)$ and $x_3(n)$ from the z-transform of $x_1(n)$.

Solution: It can be seen that $x_2(n) = x_1(n+2)$ and $x_3(n) = x_1(n-2)$

Using the shifting property we have

$$X_2(z) = z^2 X_1(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$$

and

$$X_3(z) = z^{-2} X_1(z) = z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-1}$$

Properties of the z-Transform

3. **Scaling in the z-domain**
   If

$$x(n) \overset{z}{\Longleftrightarrow} X(z) \quad \text{ROC: } r_1 < |z| < r_2$$

   then

$$a^n x(n) \overset{z}{\Longleftrightarrow} X\left(a^{-1}z\right) \quad \text{ROC: } |a|r_1 < |z| < |a|r_2$$

Example 3: Scaling in the z-domain

Given $x(n) = 2^n u(n)$

Solution:

$$u(n) \overset{z}{\Longleftrightarrow} \frac{1}{1 - z^{-1}}, \quad \text{ROC: } |z| > 1$$

$$x(n) \overset{z}{\Longleftrightarrow} \frac{1}{1 - 2z^{-1}}, \quad \text{ROC: } |z| > 2$$

Properties of the z-Transform

4. **Time-reversal**
   If

$$x(n) \overset{z}{\Longleftrightarrow} X(z) \quad \text{ROC: } r_1 < |z| < r_2$$

   then

$$x(-n) \overset{z}{\Longleftrightarrow} X\left(z^{-1}\right) \quad \text{ROC: } \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

Example 4: Time-reversal

Given $x(n) = 2^{-n}u(-n)$

Solution:

$$2^{n}u(n) \overset{z}{\Longleftrightarrow} \frac{1}{1-2z^{-1}}, \quad \text{ROC: } |z| > 2$$

$$2^{-n}u(-n) \overset{z}{\Longleftrightarrow} \frac{1}{1-2(z^{-1})^{-1}} = \frac{1}{1-2z}, \quad \text{ROC: } |z| < \frac{1}{2}$$

Properties of the z-Transform

5. **Differentiation in the z-domain**
   If

$$x(n) \overset{z}{\Longleftrightarrow} X(z) \quad \text{ROC: } r_1 < |z| < r_2$$

   then

$$nx(n) \overset{z}{\Longleftrightarrow} -z\frac{dX(z)}{dz} \quad \text{ROC: } r_1 < |z| < r_2$$

Example 5: Differentiation in the z-domain

Given $x(n) = na^n u(n)$

Solution:

$$a^n u(n) \overset{z}{\Longleftrightarrow} \frac{1}{1 - az^{-1}}, \quad \text{ROC: } |z| > |a|$$

$$na^n u(n) \overset{z}{\Longleftrightarrow} -z\frac{d}{dz}\left(\frac{1}{1 - az^{-1}}\right) = -z\left(\frac{-az^{-2}}{(1 - az^{-1})^2}\right)$$

$$= \frac{az^{-1}}{(1 - az^{-1})^2} \quad \text{ROC: } |z| > |a|$$

If $a = 1$:

$$nu(n) \overset{z}{\Longleftrightarrow} \frac{z^{-1}}{(1 - z^{-1})^2} \quad \text{ROC: } |z| > 1$$

Properties of the z-Transform

6. **Convolution of two sequences**.
   If
   $$x_1(n) \overset{z}{\Longleftrightarrow} X_1(z) \quad \text{and} \quad x_2(n) \overset{z}{\Longleftrightarrow} X_2(z)$$
   then
   $$x(n) = x_1(n) * x_2(n) \overset{z}{\Longleftrightarrow} X(z) = X_1(z)X_2(z)$$
   ROC: ROC $X_1(z) \cap$ ROC $X_2(z)$

Example 6: Convolution of two sequences

Compute the convolution $x(n)$ of the signals $x_1(n)$ and $x_2(n)$

$$x_1(n) = [\underset{\uparrow}{1}, -2, 1] \qquad x_2(n) = \begin{cases} 1, & 0 \leq n \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

Solution:

$$X_1(z) = 1 - 2z^{-1} + z^{-2}$$
$$X_2(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5}$$

Using the property, multiplication yields

$$X(z) = X_1(z)X_2(z) = 1 - z^{-1} - z^{-6} + z^{-7}$$

Hence,

$$x(n) = [\underset{\uparrow}{1}, -1, 0, 0, 0, 0, -1, 1]$$

Properties of z-Transforms

7. **Correlation of two sequences**.
   If

$$x_1(n) \overset{z}{\Longleftrightarrow} X_1(z) \quad \text{and} \quad x_2(n) \overset{z}{\Longleftrightarrow} X_2(z)$$

   then

$$r_{x_1x_2}(l) = \sum_{n=-\infty}^{\infty} x_1(n)x_2(n-l) \overset{z}{\Longleftrightarrow} R_{x_1x_2}(z) = X_1(z)X_2(z^{-1})$$

   ROC: ROC $X_1(z) \cap$ ROC $X_2(z^{-1})$

Properties of z-Transforms

8. **Multiplication of two sequences**.
   If
   $$x_1(n) \overset{z}{\Longleftrightarrow} X_1(z) \quad \text{and} \quad x_2(n) \overset{z}{\Longleftrightarrow} X_2(z)$$
   then
   $$x(n) = x_1(n)x_2(n) \overset{z}{\Longleftrightarrow} X(z) = \frac{1}{2\pi j} \oint_C X_1(\nu) X_2(\frac{z}{\nu}) \nu^{-1} d\nu$$
   ROC: If $X_1(\nu)$ converges for $r_1 < |\nu| < r_{1u}$ and $X_2(\nu)$ converges for $r_{2l} < |\nu| < r_{2u}$, then ROC of $X(z)$ is at least:
   $$r_{1l} r_{2l} < |z| < r_{1u} r_{2u}$$

Properties of z-Transforms

9. **Initial value Theorm**.
   If $x(n)$ is *causal* [i.e. $x(n) = 0$ for $n < 0$]. then

$$x(0) = \lim_{z \to \infty} X(z)$$

Common z-Transform pairs

Table: Some Common z-Transform Pairs (Part 1)

	Signal, $x(n)$	z-Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
3	$a^n u(n)$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
4	$na^n u(n)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
5	$-a^n u(-n-1)$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
6	$-na^n u(-n-1)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$

Common z-Transform pairs

Table: Some Common z-Transform Pairs (Part 2)

	Signal, $x(n)$	z-Transform, $X(z)$	ROC
7	$\cos(\omega_0 n)u(n)$	$\frac{1-z^{-1}\cos\omega_0}{1-2z^{-1}\cos\omega_0+z^{-2}}$	$\lvert z\rvert > 1$
8	$\sin(\omega_0 n)u(n)$	$\frac{z^{-1}\sin\omega_0}{1-2z^{-1}\cos\omega_0+z^{-2}}$	$\lvert z\rvert > 1$
9	$(a^n\cos(\omega_0 n))u(n)$	$\frac{1-az^{-1}\cos\omega_0}{1-2az^{-1}\cos\omega_0+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$
10	$(a^n\sin(\omega_0 n))u(n)$	$\frac{az^{-1}\sin\omega_0}{1-2az^{-1}\cos\omega_0+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$

Table of z-Transform properties

Table: Table of z-Transform Properties (Part 1)

Property	**Sequence**	**Transform**	**ROC**
Linearity	$ax_1(n) + bx_2(n)$	$aX_1(z) + bX_2(z)$	At least $ROC_1 \cap ROC_2$
Time-shift	$x(n-k)$	$z^{-k}X(z)$	$X(z) \setminus z \in \{0, \infty\}$
z-Domain Scaling	$a^n x(n)$	$X(a^{-1}z)$	$\lvert a_1 \rvert r_1 < \lvert z \rvert < \lvert a_2 \rvert r_2$
Time-reverse	$x(-n)$	$X(z^{-1})$	$\frac{1}{r_2} < \lvert z \rvert < \frac{1}{r_1}$
Derivative	$nx(n)$	$-z\frac{dX(z)}{dz}$	$r_1 < \lvert z \rvert < r_2$
Convolution	$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$	At least $ROC_1 \cap ROC_2$
Correlation	$\sum_{n=-\infty}^{\infty} x_1(n)x_2(n-l)$	$X_1(z)X_2(z^{-1})$	$ROCX_1(z) \cap ROCX_2(z)$

Table of z-Transform properties

Table: Table of z-Transform Properties (Part 2)

Property	**Sequence**	**Transform**	**ROC**
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi j}\oint_C X_1(v)X_2(\frac{z}{v})v^{-1}dv$	$r_{1l}r_{2l} < \lvert z\rvert < r_{1u}r_{2u}$
Initial Value	$x(n) = 0$ for $n < 0$	$x(0) = \lim_{z\to\infty} X(z)$	

Linear Constant Coefficient Difference Equations LCCDE

Linear Systems defined as:

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k)$$

Taking the z-transform of both sides we have:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{B(z)}{A(z)}$$

Rational z-Transforms

When $H(z)$ is a ratio of two polynomials in z or z^{-1}, then it is rational of the form:

$$H(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

If $a_0 \neq 0$ and $b_0 \neq 0$, factor out $b_0 z^{-M}$ and $a_0 z^{-N}$

$$H(z) = \frac{N(z)}{D(z)} = \frac{b_0 z^{-M}}{a_0 z^{-N}} \frac{z^M + (b_1/b_0) z^{M-1} + \cdots + (b_M/b_0)}{z^N + (a_1/a_0) z^{N-1} + \cdots + (a_N/a_0)}$$

Express $N(z)$ and $D(z)$ in factored form

$$H(z) = \frac{N(z)}{D(z)} = \frac{b_0}{a_0} z^{-M+N} \frac{(z - z_1)(z - z_2) \cdots (z - z_M)}{(z - p_1)(z - p_2) \cdots (z - p_N)}$$

Rational z-Transforms

The rational z-Transform $X(z)$ can be expressed as:

$$H(z) = \frac{b_0}{a_0} z^{N-M} \frac{\prod_{k=1}^{M}(z - z_k)}{\prod_{k=1}^{N}(z - p_k)}$$

Observations:

1. $H(z)$ has M finite ZEROS: $z = z_1, z_2, \ldots, z_M$
2. $H(z)$ has N finite POLES: $z = p_1, p_2, \ldots, p_N$
3. There are $|N - M|$ ZEROS (if $N > M$) or POLES (if $M > N$) at the origin
4. A ZERO exist at infinity if $H(\infty) = 0$
5. A POLE exist at infinity if $H(\infty) = \infty$

Rational z-Transforms

The rational z-Transform $H(z)$ can be expressed as:

$$H(z) = \frac{b_0}{a_0} z^{N-M} \frac{\prod_{k=1}^{M}(z - z_k)}{\prod_{k=1}^{N}(z - p_k)}$$

Observations:

6. $H(z)$ has exactly the same number of POLES as ZEROS (including poles and zeros at infinity)
7. $H(z)$ can be graphically represented by a Pole-Zero plot (or pattern) in the z-plane. [poles - $\times$, zeros - $\circ$]
8. ROC of a z-transform should not contain any poles
9. Given a Pole-Zero plot, we can determine $H(z)$

FIR Systems

If the coefficients of a_k is zero, we have:

$$y(n) = \sum_{k=0}^{M} b_k x(n-k)$$

The impulse response is:

$$h(n) = \begin{cases} b_k & 0 \leq n \leq M \\ 0 & \text{elsewhere} \end{cases}$$

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} = b_0 z^{-M} \prod_{k=1}^{M} (z - z_k)$$

This is the Finite Impulse Response (FIR) system. (All zeros system)

IIR Systems

When a system has both poles and zeros, $H(z)$ can be expressed as a partial fraction expansion.

$$H(z) = \sum_{k=1}^{N} \frac{A_k}{1 - p_k z^{-k}}$$

The impulse response is given as:

$$h(n) = \sum_{k=1}^{N} A_k (p_k)^n u(n)$$

Each pole contributes an exponential mode of infinite duration, IIR system.

Pole-Zero plots

Determine the Pole-Zero plot for the signal

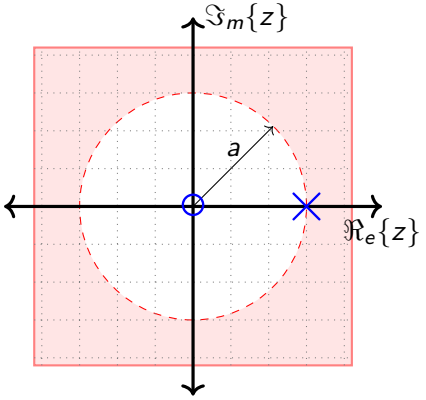
$$x(n) = a^n u(n), \quad a > 0$$

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

ROC: $|z| > a$

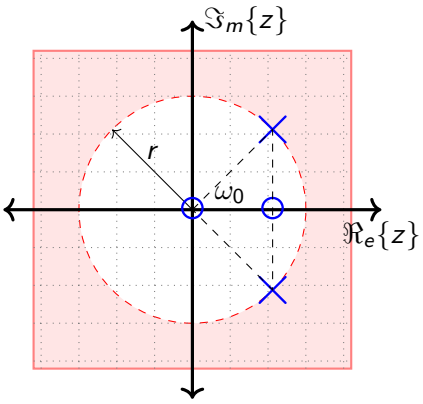
1. One ZERO at $z_1 = 0$
2. One POLE at $p_1 = a$

Note: POLE $p_1 = a$ is not included in the ROC



Pole-Zero plots

Determine the z-transform that corresponds to the P-Z plot:



1. The ZEROS are $z_1 = 0$ and $z_2 = r\cos\omega_0$
2. The POLES are $p_1 = re^{j\omega_0}$ and $p_2 = re^{-j\omega_0}$

Pole-Zero plots (continued)

From the locations of the poles and zeros we have:

$$\begin{aligned} X(z) &= G\frac{(z-z_1)(z-z_2)}{(z-p_1)(z-p_2)} \\ &= G\frac{z(z-r\cos\omega_0)}{(z-re^{j\omega_0})(z-re^{-j\omega_0})} = G\frac{z^2-rz\cos\omega_0}{z^2-2rz\cos\omega_0+r^2} \\ &= G\frac{1-rz^{-1}\cos\omega_0}{1-2rz^{-1}\cos\omega_0+r^2z^{-2}} \quad \text{ROC: } |z|>r \end{aligned}$$

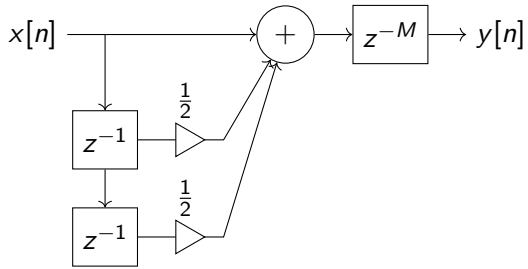
Note: If a polynomial has real coefficients, its roots are either real or occur in complex-conjugate pairs.

Stability and the z-transform

- An LTI system is BIBO stable *if and only if* the ROC of the system function includes the unit circle
- Causal LTI systems are BIBO stable and *if and only if* all the poles of the system and inside the unit circle

Block diagrams and z-Transforms

Determine the rational function $H(z) = Y(z)/X(z)$ that describes the system.



From the z-transform:

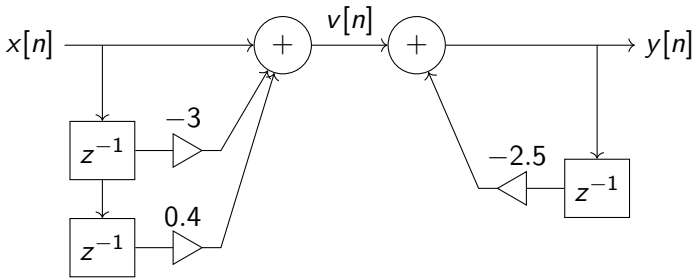
$$\left(X(z) + \tfrac{1}{2}z^{-1}X(z) + \tfrac{1}{2}z^{-2}X(z)\right) z^{-M} = Y(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = z^{-M} + \tfrac{1}{2}z^{-M-1} + \tfrac{1}{2}z^{-M-2}$$

Block diagrams and z-Transforms

Determine the block diagram, given the difference equation:

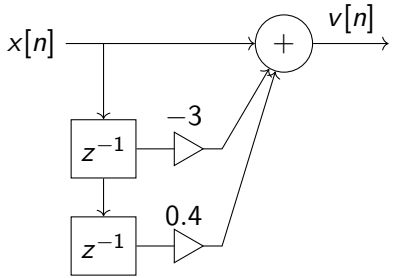
$$y[n] = x[n] - 3x[n-1] + 0.4x[n-2] - 2.5y[n-1]$$



Block diagrams and z-Transforms

Consider the left-hand diagram with input $x[n]$ and output $v[n]$:

$$v[n] = x[n] - 3x[n-1] + 0.4x[n-2]$$



Taking the z-transform:

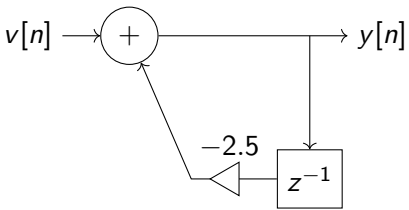
$$V(z) = X(z) - 3z^{-1}X(z) + 0.4z^{-2}X(Z)$$

$$H_1(z) = V(z)/X(z) = 1 - 3z^{-1} + 0.4z^{-2}$$

Block diagrams and z-Transforms

Consider the right-hand diagram with input $v[n]$ and output $y[n]$.

$$y[n] = v[n] - 2.5y[n-1]$$



Taking the z-transform

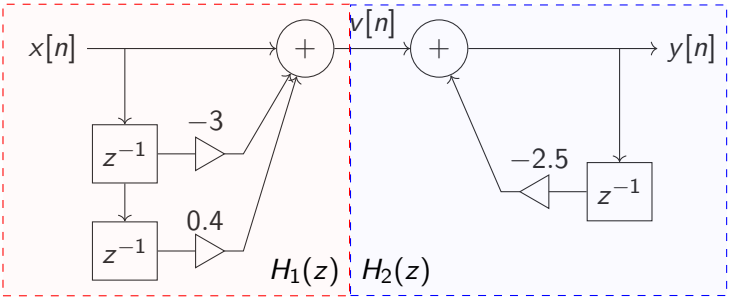
$$Y(z) = V(z) - 2.5z^{-1}Y(z)$$

$$H_2(z) = Y(z)/V(z) = \frac{1}{1 + 2.5z^{-1}}$$

Block diagrams and z-Transforms

The block diagram system function $H(z)$ is a cascade of $H_1(z)H_2(z)$.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{V(z)Y(z)}{X(z)V(z)} = H_1(z)H_2(z)$$

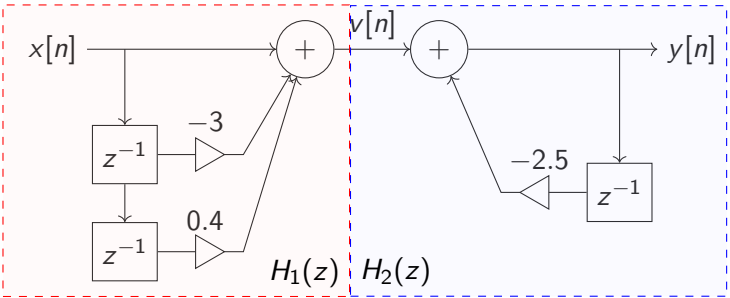


$H_1(z) = 1 - 3z^{-1} + 0.4z^{-2}$ $\quad$ $H_2(z) = \frac{1}{1+2.5z^{-1}}$

Block diagrams and z-Transforms

System function $H(z)$: Direct Form I realization

$$H(z) = H_1(z)H_2(z) = \frac{1 - 3z^{-1} + 0.4z^{-2}}{1 + 2.5z^{-1}}$$

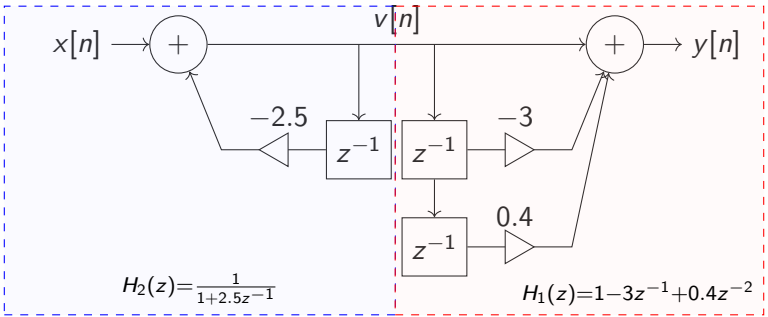


$H_1(z) = 1 - 3z^{-1} + 0.4z^{-2}$ $\quad$ $H_2(z) = \frac{1}{1+2.5z^{-1}}$

Block diagrams and z-Transforms

System function $H(z)$: Multiplication is commutative

$$H(z) = H_2(z)H_1(z) = \frac{1 - 3z^{-1} + 0.4z^{-2}}{1 + 2.5z^{-1}}$$

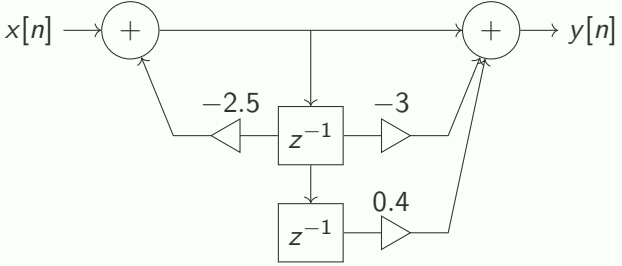


We can combine the delay elements and the node at $v[n]$.

Block diagrams and z-Transforms

System function $H(z)$: Direct Form II realization

$$H(z) = H_2(z)H_1(z) = \frac{1 - 3z^{-1} + 0.4z^{-2}}{1 + 2.5z^{-1}}$$



Direct Form I and Direct Form II

